

Biology 30 Unit A Sample Diploma-Style Exam 2

Biology 30 Unit A: Nervous and Endocrine Systems

Drawn from the attached Unit A lesson materials: Lesson 1 homeostasis and nervous-system foundations; Lesson 2 CNS integration; Lesson 3 sensory systems and perception; Lesson 4 endocrine regulation and nervous-endocrine integration.

Instructions: Select the best answer for each multiple-choice item. Answer constructed-response items using complete biological explanations. Each CR item is worth 3 marks unless otherwise directed.

Section 1 - Multiple Choice (60 questions)

MC 1. A person with impaired vasodilation during overheating would have difficulty because

- A. heat receptors would stop existing.
- B. sweating would become positive feedback automatically.
- C. the cerebellum would secrete insulin.
- D. less heat would be lost through skin blood vessels.

MC 2. Which event most directly occurs at threshold in an axon?

- A. Neurotransmitter is released before the action potential starts.
- B. The membrane potential remains permanently at rest.
- C. Voltage-gated sodium channels open and rapid depolarization begins.
- D. Hormones enter the axon through receptors.

MC 3. Why do action potentials usually travel one direction along an axon?

- A. Recently depolarized regions enter a refractory period.
- B. Neurotransmitters pull the impulse backward.
- C. The nucleus pushes impulses away from dendrites.
- D. The myelin sheath eliminates all ion movement.

MC 4. In a neuromuscular junction, acetylcholine is removed or broken down mainly to

- A. make each action potential larger.
- B. convert muscle contraction into a hormone response.
- C. stop continuous stimulation of the muscle fibre.
- D. prevent sodium from leaving the blood.

MC 5. A toxin that prevents neurotransmitter release from a motor neuron would most directly

- A. increase hormone receptor numbers in the liver.
- B. make sensory receptors detect ultraviolet light.
- C. reduce stimulation of the target muscle.
- D. strengthen spinal cord protection by meninges.

MC 6. Which statement best distinguishes dendrites from axons?

- A. Dendrites generally receive information; axons conduct impulses away from the cell body.
- B. Dendrites release hormones into blood; axons store DNA.
- C. Dendrites form myelin; axons are endocrine cells.
- D. Dendrites are found only in glands; axons are found only in muscles.

MC 7. A person has normal receptor detection but poor interpretation of touch location. Which level is most likely affected?

- A. The stimulus itself.
- B. The endocrine glands only.
- C. The effector muscle only.
- D. Central processing in the CNS.

MC 8. Why does CNS coordination matter during cold exposure?

- A. Multiple effectors such as skeletal muscle, blood vessels, and behaviour must be coordinated.
- B. Only one organ can respond to any stimulus.
- C. Receptors decide which response is best without control centres.
- D. Cold exposure is controlled entirely by digestion.

MC 9. Which finding would most directly suggest cerebellar damage?

- A. Inability of the pancreas to release insulin.
- B. Inability of hair cells to transduce sound.
- C. Complete lack of receptors in the skin.
- D. Poor balance and inaccurate movement coordination.

MC 10. Which brain region would be most directly involved in forming new memories from experience?

- A. Pancreas.
- B. Hippocampus.
- C. Retina.
- D. Adrenal cortex.

MC 11. The blood-brain barrier is important because it

- A. helps regulate which substances enter nervous tissue from blood.
- B. prevents all hormones from existing.
- C. causes action potentials to travel both directions.
- D. replaces the skull as mechanical protection.

MC 12. A patient can detect sound but cannot understand speech meaning. This suggests

- A. hair cells are always absent.
- B. the pancreas is overproducing glucagon.
- C. central interpretation may be impaired even if receptors function.
- D. all sensory input bypasses the brain.

MC 13. Which statement about pain perception is most accurate?

- A. Pain depends on receptor input and CNS interpretation.
- B. Pain always exactly equals tissue damage severity.
- C. Pain occurs only when hormones bind receptors.
- D. Pain is generated directly by skeletal muscles without nerves.

MC 14. Why can two people perceive the same stimulus differently?

- A. Action potentials are larger in one person.
- B. Stimuli become different chemical elements in each person.
- C. Perception is influenced by attention, prior experience, and neural processing.
- D. Receptors make conscious decisions.

MC 15. Proprioceptors are especially important for

- A. detecting blood glucose in the pancreas.
- B. secreting TSH from the pituitary.
- C. converting light into colour only.
- D. monitoring body position and movement.

MC 16. Which example best shows sensory transduction?

- A. Hair cells converting vibration into neural signals.
- B. The adrenal gland releasing cortisol into blood.
- C. A muscle contracting after receiving a motor signal.
- D. The liver storing glucose as glycogen.

MC 17. Which disorder would most directly impair focusing light precisely on the retina?

- A. Diabetes mellitus.
- B. Reduced glucagon release.
- C. Damage to adrenal cortex only.
- D. Myopia or hyperopia.

MC 18. Why is central sensory filtering adaptive?

- A. It prevents receptors from detecting any stimulus.
- B. It ensures all input reaches consciousness equally.
- C. It allows relevant signals to receive attention while background input is reduced.
- D. It changes hormones into action potentials.

MC 19. If a target cell lacks receptors for ADH, which result is most likely?

- A. ADH will become a nerve impulse.
- B. The cell will not respond normally even if ADH is present in blood.
- C. The cell will respond more strongly than receptor-bearing cells.
- D. The hypothalamus will no longer receive any sensory input.

MC 20. Which gland pair is most directly involved in insulin and glucagon regulation?

- A. Pancreatic islets.
- B. Thyroid follicles.
- C. Parathyroid glands.
- D. Adrenal medulla only.

MC 21. Low blood calcium most directly stimulates secretion of

- A. insulin from pancreatic beta cells.
- B. melatonin from the thyroid gland.
- C. PTH from the parathyroid glands.
- D. acetylcholine from the adrenal cortex.

MC 22. High metabolic rate, heat intolerance, and weight loss could be consistent with excessive

- A. PTH deficiency only.
- B. blocked spinal reflexes only.
- C. thyroid hormone activity.
- D. hair-cell damage in the cochlea.

MC 23. A tropic hormone is best described as one that

- A. stimulates another endocrine gland to release hormone.
- B. contracts skeletal muscle directly.
- C. is a sensory receptor in skin.
- D. makes action potentials larger.

MC 24. Which pair correctly matches hormone and general effect?

- A. Insulin - increased blood glucose release by liver.
- B. Glucagon - lowered blood glucose after meals.
- C. ADH - increased water reabsorption by kidneys.
- D. PTH - decreased blood calcium when levels are low.

MC 25. Why is feedback failure dangerous in endocrine pathways?

- A. It prevents hormones from entering the bloodstream at all times.
- B. It affects only one cell with no downstream effects.
- C. It immediately turns all endocrine pathways into reflex arcs.
- D. It can create compounding, system-wide imbalance over time.

MC 26. Which example requires both nervous and endocrine coordination?

- A. A single action potential in one axon only.
- B. A receptor detecting one light stimulus with no response.
- C. A prolonged stress response with immediate alertness and sustained energy mobilization.
- D. A static anatomical description of the skull.

MC 27. Why are hormones described as messages rather than commands?

- A. Every hormone forces every cell to respond identically.
- B. Target-cell response depends on receptor presence and intracellular state.
- C. Hormones do not need target cells.
- D. Hormones never influence physiology.

MC 28. A patient has normal TSH but low target tissue response to thyroid hormone. Which explanation is plausible?

- A. Receptor or intracellular response in target cells may be impaired.
- B. The hormone must be absent from blood.
- C. The retina must be damaged.
- D. Spinal reflexes must control metabolism alone.

MC 29. Endocrine control alone would be dangerous in emergencies because

- A. hormones cannot influence any organs.
- B. blood-borne hormone transport is often too slow for immediate protective action.
- C. endocrine glands lack all receptors.
- D. hormones act only before birth.

MC 30. Nervous control alone would be limiting for growth and metabolism because

- A. neurons cannot communicate with any gland.
- B. the nervous system lacks feedback in all pathways.
- C. action potentials are slower than all hormones.
- D. neural signals are brief and localized compared with sustained hormonal regulation.

MC 31. Which statement best identifies an effector in a homeostatic pathway?

- A. A structure that detects the initial change only.
- B. An organ or tissue that carries out the response directed by the control centre.
- C. A molecule that stores genetic information.
- D. A brain region that always causes positive feedback.

MC 32. During dehydration, increased ADH secretion helps homeostasis by

- A. increasing water loss in urine.
- B. reducing kidney response to blood osmolarity.
- C. causing the liver to release digestive enzymes.
- D. increasing water reabsorption by the kidneys.

MC 33. Which condition would most directly trigger glucagon release?

- A. Rising blood glucose after a meal.
- B. High calcium concentration in blood.
- C. Falling blood glucose between meals.
- D. Bright light entering the retina.

MC 34. Which condition would most directly trigger insulin release?

- A. Falling blood glucose during fasting.
- B. Rising blood glucose after a meal.
- C. Low calcium concentration in blood.
- D. A withdrawal reflex from heat.

MC 35. Which statement best explains why the pancreas is an endocrine gland in glucose regulation?

- A. It sends action potentials directly to every muscle.
- B. It releases hormones into blood that influence distant target tissues.
- C. It detects only external stimuli.
- D. It is part of the spinal cord.

MC 36. A hormone binds to its receptor but the cell response is weak. The best conclusion is that

- A. transport failure must be the only cause.
- B. the hormone cannot be endocrine.
- C. negative feedback has been eliminated in all systems.
- D. a problem may exist in intracellular signaling or downstream effector mechanisms.

MC 37. Which example is most clearly positive feedback?

- A. Insulin lowering blood glucose after a meal.
- B. Sweating lowering body temperature.
- C. Oxytocin increasing uterine contractions during labour.
- D. ADH increasing water retention during dehydration.

MC 38. Which statement best describes the role of the pituitary gland?

- A. It acts independently of the hypothalamus.
- B. It often amplifies hypothalamic signals by releasing hormones that regulate other glands.
- C. It is the only receptor in the endocrine system.
- D. It replaces all nervous-system functions.

MC 39. Which hormone abbreviation matches thyroid-stimulating hormone?

- A. ADH.
- B. PTH.
- C. hGH.
- D. TSH.

MC 40. Which hormone abbreviation matches antidiuretic hormone?

- A. ADH.
- B. ACTH.
- C. LH.
- D. PRL.

MC 41. Which hormone abbreviation matches parathyroid hormone?

- A. FSH.
- B. hCG.
- C. TSH.
- D. PTH.

MC 42. Which hormone abbreviation matches adrenocorticotrophic hormone?

- A. ADH.
- B. GnRH.
- C. ACTH.
- D. PTH.

MC 43. Which response is most directly associated with the sympathetic division?

- A. Increased digestion during rest.
- B. Decreased alertness during danger.
- C. Direct production of insulin by neurons.
- D. Increased heart rate and mobilization of energy during threat.

MC 44. Which response is most directly associated with the parasympathetic division?

- A. Increased heart rate during threat.
- B. Increased digestive activity during rest.
- C. Increased cortisol secretion from the retina.
- D. Blocked cerebrospinal fluid production.

MC 45. Which statement correctly compares receptors and control centres?

- A. Receptors detect change; control centres interpret input and coordinate response.
- B. Receptors carry out responses; control centres detect only external stimuli.
- C. Receptors are always glands; control centres are always muscles.
- D. Receptors and control centres are identical in every pathway.

MC 46. Which observation best supports all-or-none action potentials?

- A. Stronger stimuli always make individual action potentials taller.
- B. Action potentials are hormones that vary continuously in size.
- C. A threshold stimulus produces a full action potential, while a subthreshold stimulus does not.
- D. Every stimulus produces the same conscious perception.

MC 47. During repolarization, the membrane potential moves back toward resting mainly because

- A. sodium ions flood into the neuron continuously.
- B. neurotransmitters enter the nucleus.
- C. the pituitary releases TSH.
- D. potassium ions leave the neuron through voltage-gated channels.

MC 48. The sodium-potassium pump is important after neural activity because it

- A. creates neurotransmitters in the synaptic cleft.
- B. helps maintain ion gradients needed for future impulses.
- C. converts hormones into electrical impulses.
- D. digests all receptor proteins.

MC 49. Which structure receives neurotransmitter at a chemical synapse?

- A. Receptor proteins on the postsynaptic membrane.
- B. DNA codons inside the axon.
- C. Cerebrospinal fluid only.
- D. Insulin receptors on pancreatic beta cells only.

MC 50. Which example best illustrates CNS plasticity?

- A. The brain learning to interpret new patterns of input from a cochlear implant.
- B. The pancreas releasing insulin after every meal.
- C. The skull protecting the brain from impact.
- D. A receptor detecting one unchanging stimulus forever.

MC 51. Why can artificial sensory technologies have limitations?

- A. They always regenerate original receptors perfectly.
- B. They eliminate the need for brain interpretation.
- C. They convert all sounds into blood glucose.
- D. They may not reproduce the full precision and adaptability of natural receptor systems.

MC 52. A vestibular disorder would most likely affect

- A. blood calcium homeostasis.
- B. balance, spatial orientation, and nausea during motion.
- C. insulin receptor binding.
- D. DNA replication in skin cells.

MC 53. Which role is most closely associated with the retina?

- A. Filtering blood plasma.
- B. Transducing light stimuli into neural signals.
- C. Secreting PTH.
- D. Producing cerebrospinal fluid.

MC 54. Which role is most closely associated with the organ of Corti?

- A. Regulating blood glucose with insulin.
- B. Coordinating uterine contractions.
- C. Protecting the brain with bone.
- D. Transducing sound vibration through hair cells.

MC 55. Which best explains why receptors are specialized?

- A. All receptors detect all stimuli equally.
- B. Different receptor proteins and structures respond to particular forms of energy or molecules.
- C. Receptors are effectors that always contract.
- D. Receptors respond only to hormones.

MC 56. Which statement most accurately compares endocrine and nervous communication distance?

- A. Endocrine hormones travel through blood broadly; neurons communicate along specific pathways and synapses.
- B. Hormones travel only across synaptic clefts.
- C. Neurons communicate by releasing hormones into all blood vessels only.
- D. Both systems use identical routes and timing.

MC 57. Which effect would likely follow adrenal medulla activation?

- A. Rapid release of epinephrine to support fight-or-flight responses.
- B. Release of insulin to lower blood glucose only.
- C. Reduced alertness during danger.
- D. Regeneration of cochlear hair cells.

MC 58. Which condition would suggest impaired negative feedback in a hormone pathway?

- A. A stimulus is corrected and hormone secretion decreases.
- B. A target hormone remains high because upstream stimulation is not reduced.
- C. A receptor detects change and a control centre adjusts output.
- D. A reflex arc withdraws a hand from heat.

MC 59. Which statement best explains why homeostasis requires communication among systems?

- A. Each organ responds independently with no shared signals.
- B. Cells and organs must coordinate responses to maintain stable internal conditions.
- C. Only voluntary muscles participate in homeostasis.
- D. Stable conditions occur without receptors or effectors.

MC 60. Which question is most aligned with Biology 30 Unit A?

- A. How does meiosis produce genetic variation in gametes?
- B. How do predator-prey cycles affect carrying capacity?
- C. How do nervous and endocrine systems coordinate physiological processes to maintain homeostasis?
- D. How are restriction enzymes used to cut DNA?

Section 2 - Constructed Response (12 questions)

CR 1. A person exposed to cold begins shivering, reduces blood flow to the skin, and seeks shelter.

- Identify the role of the CNS in coordinating this response.
- Explain why more than one effector is useful.
- Explain why independent organ responses would be less effective.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 2. A patient's eyes detect light normally, but visual meaning cannot be interpreted.

- Identify whether this is mainly a receptor-level or CNS-level problem.
- Explain the role of the occipital lobe in this case.
- Explain why detection alone is not the same as perception.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 3. A patient has a damaged cerebellum but intact skeletal muscles.

- Predict one effect on movement.
- Explain why intact muscles are not enough for normal movement.
- Explain how this illustrates structure-function relationships in the CNS.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 4. A student says sensory systems simply record reality and send it unchanged to the brain.

- Explain why the claim is inaccurate.
- Describe the role of receptor specificity.
- Explain why perception is interpretation rather than recording.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 5. A person stops noticing a watch on the wrist after several minutes.

- Identify the process involved.
- Explain why the response to a constant stimulus decreases.
- Explain how this process can improve survival.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 6. A patient has receptor damage in the retina in one region but normal brain function.

- Predict the effect on visual input.
- Explain why the brain can compensate only partly.
- Explain what this shows about the limits of central processing.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 7. A patient responds immediately to a threat with alertness but cannot maintain physiological readiness over time.

- Explain which part of the stress response is still functioning.
- Explain which part is likely impaired.
- Explain why this demonstrates nervous-endocrine integration.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 8. A patient has normal hormone levels but weak target-tissue response.

- a. Explain why this does not rule out endocrine disease.
- b. Identify one possible target-cell cause.
- c. Explain how this supports the idea that hormones are messages, not commands.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 9. A student says the pituitary gland controls everything in the endocrine system.

- a. Explain why this is incomplete.
- b. Describe the role of the hypothalamus.
- c. Explain why the pituitary can be described as an amplifier.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 10. Compare the thyroid and parathyroid glands.

- a. State the main homeostatic role of each gland.
- b. Explain why confusing them leads to faulty reasoning.
- c. Explain how each contributes to survival.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 11. Insulin and glucagon are often described as opposite hormones.

- a. Explain how each affects blood glucose.
- b. Explain why it is misleading to think of them as unrelated.
- c. Explain why both are needed for homeostasis.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

CR 12. A target hormone normally feeds back to inhibit hypothalamic and pituitary signals, but this inhibition fails.

- a. Describe what normally happens in negative feedback.
- b. Predict what may happen when inhibition fails.
- c. Explain why effects can spread beyond one gland.

Scoring reminder: 3 marks total - 1 mark for accurate concept, 1 mark for reasoning, 1 mark for correct Biology 30 terminology.

Section 3 - Answer Key and Scoring Guide

Multiple Choice Answer Key

1	D	16	A	31	B	46	C
2	C	17	D	32	D	47	D
3	A	18	C	33	C	48	B
4	C	19	B	34	B	49	A
5	C	20	A	35	B	50	A
6	A	21	C	36	D	51	D
7	D	22	C	37	C	52	B
8	A	23	A	38	B	53	B
9	D	24	C	39	D	54	D
10	B	25	D	40	A	55	B
11	A	26	C	41	D	56	A
12	C	27	B	42	C	57	A
13	A	28	A	43	D	58	B
14	C	29	B	44	B	59	B
15	D	30	D	45	A	60	C

MC Rationales

1. **D.** less heat would be lost through skin blood vessels.
2. **C.** Voltage-gated sodium channels open and rapid depolarization begins.
3. **A.** Recently depolarized regions enter a refractory period.
4. **C.** stop continuous stimulation of the muscle fibre.
5. **C.** reduce stimulation of the target muscle.
6. **A.** Dendrites generally receive information; axons conduct impulses away from the cell body.
7. **D.** Central processing in the CNS.
8. **A.** Multiple effectors such as skeletal muscle, blood vessels, and behaviour must be coordinated.
9. **D.** Poor balance and inaccurate movement coordination.
10. **B.** Hippocampus.
11. **A.** helps regulate which substances enter nervous tissue from blood.
12. **C.** central interpretation may be impaired even if receptors function.
13. **A.** Pain depends on receptor input and CNS interpretation.
14. **C.** Perception is influenced by attention, prior experience, and neural processing.
15. **D.** monitoring body position and movement.
16. **A.** Hair cells converting vibration into neural signals.
17. **D.** Myopia or hyperopia.
18. **C.** It allows relevant signals to receive attention while background input is reduced.
19. **B.** The cell will not respond normally even if ADH is present in blood.
20. **A.** Pancreatic islets.
21. **C.** PTH from the parathyroid glands.
22. **C.** thyroid hormone activity.
23. **A.** stimulates another endocrine gland to release hormone.
24. **C.** ADH - increased water reabsorption by kidneys.
25. **D.** It can create compounding, system-wide imbalance over time.

- 26. **C.** A prolonged stress response with immediate alertness and sustained energy mobilization.
- 27. **B.** Target-cell response depends on receptor presence and intracellular state.
- 28. **A.** Receptor or intracellular response in target cells may be impaired.
- 29. **B.** blood-borne hormone transport is often too slow for immediate protective action.
- 30. **D.** neural signals are brief and localized compared with sustained hormonal regulation.
- 31. **B.** An organ or tissue that carries out the response directed by the control centre.
- 32. **D.** increasing water reabsorption by the kidneys.
- 33. **C.** Falling blood glucose between meals.
- 34. **B.** Rising blood glucose after a meal.
- 35. **B.** It releases hormones into blood that influence distant target tissues.
- 36. **D.** a problem may exist in intracellular signaling or downstream effector mechanisms.
- 37. **C.** Oxytocin increasing uterine contractions during labour.
- 38. **B.** It often amplifies hypothalamic signals by releasing hormones that regulate other glands.
- 39. **D.** TSH.
- 40. **A.** ADH.
- 41. **D.** PTH.
- 42. **C.** ACTH.
- 43. **D.** Increased heart rate and mobilization of energy during threat.
- 44. **B.** Increased digestive activity during rest.
- 45. **A.** Receptors detect change; control centres interpret input and coordinate response.
- 46. **C.** A threshold stimulus produces a full action potential, while a subthreshold stimulus does not.
- 47. **D.** potassium ions leave the neuron through voltage-gated channels.
- 48. **B.** helps maintain ion gradients needed for future impulses.
- 49. **A.** Receptor proteins on the postsynaptic membrane.
- 50. **A.** The brain learning to interpret new patterns of input from a cochlear implant.
- 51. **D.** They may not reproduce the full precision and adaptability of natural receptor systems.
- 52. **B.** balance, spatial orientation, and nausea during motion.
- 53. **B.** Transducing light stimuli into neural signals.
- 54. **D.** Transducing sound vibration through hair cells.
- 55. **B.** Different receptor proteins and structures respond to particular forms of energy or molecules.
- 56. **A.** Endocrine hormones travel through blood broadly; neurons communicate along specific pathways and synapses.
- 57. **A.** Rapid release of epinephrine to support fight-or-flight responses.
- 58. **B.** A target hormone remains high because upstream stimulation is not reduced.
- 59. **B.** Cells and organs must coordinate responses to maintain stable internal conditions.
- 60. **C.** How do nervous and endocrine systems coordinate physiological processes to maintain homeostasis?

Constructed Response Model Answers and Rubrics

CR 1. A person exposed to cold begins shivering, reduces blood flow to the skin, and seeks shelter.

Model answer: a. The CNS receives temperature information, interprets it, and coordinates responses that conserve or generate heat. b. Shivering generates heat, vasoconstriction reduces heat loss, and behaviour changes exposure, so multiple effectors address the same problem in different ways. c. Independent organs could respond inconsistently or too slowly; CNS integration produces a coordinated whole-body response.

Required concepts: CNS integration; cold stress; shivering; vasoconstriction; behaviour.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 2. A patient's eyes detect light normally, but visual meaning cannot be interpreted.

Model answer: a. This is mainly a CNS-level problem if photoreceptors and optic pathways detect light normally. b. The occipital lobe is involved in visual processing and interpretation. c. Detection provides neural input, while perception requires brain interpretation of the pattern of impulses.

Required concepts: Photoreceptors; occipital lobe; detection vs perception.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 3. A patient has a damaged cerebellum but intact skeletal muscles.

Model answer: a. The patient may show poor balance, tremor, or inaccurate coordination. b. Muscles can contract, but the cerebellum refines timing, coordination, and balance. c. Different CNS regions have specialized functions; damage to a coordinating region disrupts movement quality even if effectors still work.

Required concepts: Cerebellum; coordination; balance; effector function.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 4. A student says sensory systems simply record reality and send it unchanged to the brain.

Model answer: a. Sensory systems detect only certain stimuli and convert them into neural signals. b. Receptors are specialized for particular forms of energy or molecules and ignore others. c. The brain receives patterns of impulses, not the original stimulus, and constructs meaning from those patterns.

Required concepts: Receptor specificity; transduction; sensory filtering; perception.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 5. A person stops noticing a watch on the wrist after several minutes.

Model answer: a. The process is sensory adaptation. b. Neural response to an unchanging background stimulus decreases, so constant information is not emphasized. c. This improves survival by helping the nervous system prioritize new or changing stimuli that may require action.

Required concepts: Sensory adaptation; background stimuli; attention; survival value.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 6. A patient has receptor damage in the retina in one region but normal brain function.

Model answer: a. Input from that retinal region would be reduced or absent. b. The brain may use surrounding information or experience to interpret the scene, but it cannot fully reconstruct missing receptor input. c. Central processing depends on available sensory signals; interpretation cannot perfectly replace absent transduction.

Required concepts: Retina; photoreceptor damage; missing input; compensation limits.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 7. A patient responds immediately to a threat with alertness but cannot maintain physiological readiness over time.

Model answer: a. The immediate neural component, likely sympathetic activation, is functioning. b. The slower endocrine support of the stress response is likely impaired, reducing sustained energy mobilization. c. A normal stress response begins with fast neural signaling and is maintained by endocrine hormones, so both systems must integrate.

Required concepts: Sympathetic response; endocrine support; stress; integration.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 8. A patient has normal hormone levels but weak target-tissue response.

Model answer: a. Endocrine disease can result from receptor or signaling failure, not only abnormal hormone concentration. b. The target cells may have fewer receptors, defective receptors, or impaired intracellular signaling. c. A hormone message produces a response only if the target cell can receive and process it.

Required concepts: Receptor resistance; target cell; intracellular signaling; hormone message.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 9. A student says the pituitary gland controls everything in the endocrine system.

Model answer: a. The pituitary does not act independently; it is regulated by hypothalamic signals and feedback. b. The hypothalamus monitors conditions and links neural information to endocrine control by regulating pituitary hormone release. c. The pituitary can amplify hypothalamic signals by releasing hormones that stimulate peripheral endocrine glands.

Required concepts: Hypothalamus; pituitary; endocrine hierarchy; tropic hormones.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 10. Compare the thyroid and parathyroid glands.

Model answer: a. The thyroid mainly regulates metabolic rate, growth, and energy use; the parathyroid glands regulate blood calcium levels through PTH. b. Confusing them is faulty because they regulate different variables with different hormones and symptoms. c. Thyroid hormones support metabolism and development, while PTH maintains calcium needed for bones, muscle contraction, and nerve signaling.

Required concepts: Thyroid; parathyroid; metabolism; calcium; PTH.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 11. Insulin and glucagon are often described as opposite hormones.

Model answer: a. Insulin lowers blood glucose by promoting uptake and storage, while glucagon raises blood glucose by promoting release from stores. b. They are not unrelated because they are coordinated parts of the same regulatory system. c. Both are needed to prevent blood glucose from becoming too high after meals or too low during fasting.

Required concepts: Insulin; glucagon; blood glucose; coordinated negative feedback.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.

CR 12. A target hormone normally feeds back to inhibit hypothalamic and pituitary signals, but this inhibition fails.

Model answer: a. Normally, the target hormone reduces upstream stimulation to prevent overproduction. b. If inhibition fails, hormone levels can remain high or continue rising, pushing physiology away from the normal range. c. Effects can spread because endocrine glands and target tissues operate in integrated networks with multiple downstream effects.

Required concepts: Negative feedback; hypothalamus; pituitary; cascading endocrine imbalance.

Rubric: 1 point accurate biological concept; 1 point clear reasoning, explanation, prediction, or justification; 1 point correct Biology 30 terminology and notation.